

MAT1106 Design and Analysis of Experiments

Analysis of Variance

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Analysis of variance

Analysis of variance (ANOVA) is the most commonly used technique to test differences between two or more means

One-way ANOVA

One-way ANOVA is the simplest form of analysis of variance, where variation of a single factor is studied between groups of measurements. It is an extension of the unpaired t test and can be used to compare any number of groups of measurements.

In an unpaired t test, the test statistic is computed by dividing the difference between the sample means by the standard error of the difference. The standard error of the difference is an estimate of the variability within each group (assumed to be the same). In other words, the difference (or variability) between the samples is compared with the variability within the samples.

In one-way ANOVA, the same principle is used, with variances rather than standard deviations being used to measure variability.

Formally, if the mean of the i th group is denoted by $\bar{x}_i$, and the mean across the groups (the overall mean) is denoted by $\hat{x}$, the deviation of an individual observation x_{ij} from the overall mean can be represented as

$$x_{ij} - \hat{x} = (\bar{x}_i - \hat{x}) + (x_{ij} - \bar{x}_i) \quad (1)$$

The first term on the right-hand side $(\bar{x}_i - \hat{x})$ represents the deviation of a group mean from the overall mean and is an indication of between-group variability.

The second term on the right-hand side $(x_{ij} - \bar{x}_i)$ represents the deviation of an individual observation from the group mean for that observation and is an indication of within-group variability.

If both sides of Equation 1 are squared and the squared deviations are summed over all observations over all groups, then the following relationship is obtained:

$$\sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \hat{x})^2 = \sum_{i=1}^k \sum_{j=1}^{n_i} (\bar{x}_i - \hat{x})^2 + \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2 \quad (2)$$

The term $\sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \hat{x})^2$ is called the Total Sum of Squares.

The term $\sum_{i=1}^k \sum_{j=1}^{n_i} (\bar{x}_i - \hat{x})^2$ is called the Between Sum of Squares.

The term $\sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2$ is called the Within Sum of Squares.

Thus, the relation in Equation 2 can be written as:

$$\text{Total SS} = \text{Between SS} + \text{Within SS} \quad (3)$$

The degrees of freedom (df) for these sums of squares are as follows:

$$\text{Total } df = n - 1$$

$$\text{Between groups } df = k - 1$$

$$\text{Within groups } df = (n - 1) - (k - 1) = n - k$$

where n is the total number of observations and k is the number of groups

The significance test is based on the ratio of the Between Mean Square to the Within Mean Square, which are defined as follows:

Between mean square (Between MS) = $\text{Between SS} / (k - 1)$

Within mean square (Within MS) = $\text{Within SS} / (n - k)$

If this ratio is large, the null hypothesis is rejected; if it is small, the null hypothesis is accepted (or failed to be rejected).

Furthermore, under the null hypothesis, the ratio of Between MS to Within MS follows an F distribution with $k - 1$ and $n - k$ degrees of freedom.

The P value is obtained by comparison of the test statistic with the F distribution with $k - 1$ and $n - k$ degrees of freedom.

The results from ANOVA are typically displayed as follows:

Source of variation	df	SS	MS	F	P
Between	$k - 1$	$\sum_{i=1}^k \sum_{j=1}^{n_i} (\bar{x}_i - \bar{\hat{x}})^2$	$\frac{\text{Between SS}}{k - 1}$	$\frac{\text{Between SS}/(k-1)}{\text{Within SS}/(n-k)}$	$Pr(F_{k-1, n-k} > F)$
Within	$n - k$	$\sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2$	$\frac{\text{Within SS}}{n - k}$		
Total	$n - 1$	Between SS + Within SS			

Assumptions

The underlying assumptions for one-way analysis of variance are that the observations are independent and randomly selected from normal populations with equal variances. It is not necessary to have equal sample sizes.

If the assumptions are not met, it may be possible to transform the data.

Relationship between one-way ANOVA and unpaired t test

Since an ANOVA and an unpaired t test can both test the difference between two means, the results obtained using both techniques will be the same. When there are only two groups, the following relationship between F and t will hold:

$$F(1, dfd) = t^2(df) \quad (4)$$

where dfd is the degrees of freedom for the denominator of the F test and df is the degrees of freedom for the t test. dfd will equal df .

MAT1106 Design and Analysis of Experiments

Correlation and Regression Analysis

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Correlation and regression

The most commonly used techniques for investigating the relationship between two quantitative variables are correlation and linear regression.

Correlation quantifies the strength of the linear relationship between a pair of variables, whereas regression expresses the relationship in the form of an equation.

Correlation

When investigating a relationship between two variables, data values can be plotted to obtain a scatter diagram.

Although a scatter diagram provides a pictorial understanding of the relationship between two variables, it fails to provide any numerical relationship.

To quantify the strength of the relationship, the correlation coefficient can be calculated.

Pearson correlation coefficient

The Pearson correlation coefficient (denoted by r) is the most commonly used measure of correlation.

If there are two variables x and y , and the data take the form of n pairs (i.e., $[x_1, y_1], [x_2, y_2], \dots [x_n, y_n]$), then the correlation coefficient is given by the following equation:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} \quad (1)$$

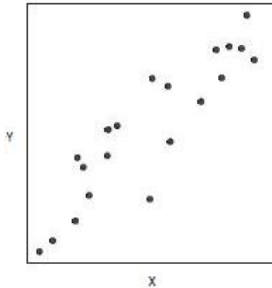
where $\bar{x}$ is the mean of the x values, and $\bar{y}$ is the mean of the y values.

The value of r lies between -1 and $+1$.

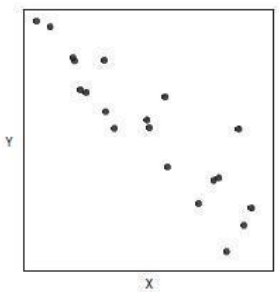
A value of the correlation coefficient close to $+1$ indicates a strong positive linear relationship.

A value close to -1 indicates a strong negative linear relationship.

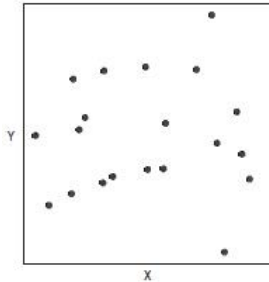
A value close to 0 indicates no linear relationship; however, there could be a nonlinear relationship between the variables.



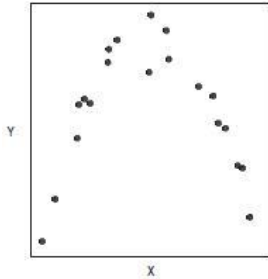
Positive linear relationship ($r = +0.9$)



Negative linear relationship ($r = -0.9$)



No relationship
($r = 0.04$)



Nonlinear relationship ($r = -0.03$)

Spearman rank correlation coefficient

The Pearson correlation coefficient cannot be used in cases where direct quantitative measurement of variables is not possible. In such situations, the Spearman rank correlation coefficient can be used.

The Spearman rank correlation coefficient (denoted by ρ) is a non-parametric measure of statistical dependence between two variables. It is defined as the Pearson correlation coefficient between the ranked variables.

For a sample of size n , the n raw scores x_i, y_i are converted to ranks, and ρ is computed using the equation for computing r .

Identical values (rank ties or value duplicates) are assigned a rank equal to the average of their positions in the ascending order of the values.

If duplicate values (ties) are known to be absent, ρ can be calculated as follows:

$$\rho = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)} \quad (2)$$

where d_i is the difference between the ranks of a variable, and n is the number of pairs ranked.

The value of ρ lies between -1 and $+1$.

The sign of ρ indicates the direction of association between x and y . If y tends to increase when x increases, ρ is positive. If y tends to decrease when x increases, ρ is negative.

A rank correlation coefficient of zero indicates that there is no tendency for y to either increase or decrease when x increases.

Correlation and causation

The presence of correlation between two variables does not necessarily imply the existence of a direct causation, though causation will always result in correlation. In general, correlation may be due to any of the following factors:

- One variable being the cause of the other variable

- Both variables being the result of a common cause

- Chance

Misinterpretations of correlation

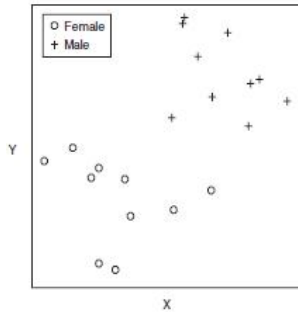
One of the most common errors in interpreting the correlation coefficient is failure to consider that there may be a third variable related to both of the variables being investigated, which is responsible for the apparent correlation.

Correlation does not necessarily imply causation.

To strengthen the case for causality, consideration must be given to other possible underlying variables.

A nonlinear relationship may exist between two variables that would be inadequately described, or possibly even undetected, by the correlation coefficient.

A data set may sometimes comprise distinct subgroups, which could result in clusters of points leading to an inflated correlation coefficient. A single outlier may produce a similar effect.



All data: $r = 0.57$; males: $r = -0.41$; females: $r = -0.26$

It is important that the values of one variable are not determined in advance or restricted to a certain range. This may lead to an invalid estimate of the true correlation coefficient because the variables are not a random sample.

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Correlation and Regression Analysis

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Regression

Regression expresses the relationship between a pair of variables in the form of an equation, usually that of a straight line (linear regression).

In simple linear regression, the effect of one variable (the predictor or x variable) on the other variable (the response or y variable) is measured. The values of y when plotted as a function of x form a straight line.

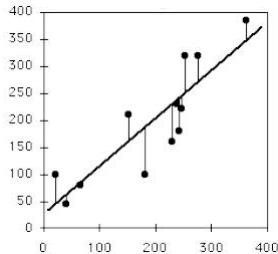
Linear regression consists of finding the best-fitting straight line through the points. The best-fitting line is called a regression line.

Method of least squares

Linear regression finds the line that best fits the data points. In this case, the best fit is defined as the line that minimizes the squared vertical distances between the data points and the line.

For a data point with an x value of x_1 and a y value of y_1 , the difference between y_1 and the value of y on the line at x_1 is calculated and then squared. This squared deviate is calculated for each data point, and the sum of these squared deviates measures how well a line fits the data.

The regression line is the one for which this sum of squared deviates is smallest.



The regression line is the line that minimizes the sum of the squared vertical distances between the points and the line

The equation for the regression line is usually expressed as $y = (\text{slope} \times x) + \text{intercept}$.

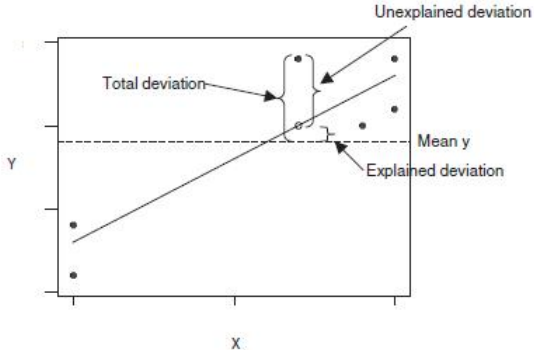
This equation can be used to predict the value of y for a given value of x and vice versa.

These predictions are best done within the range of x and y values observed in the data (interpolation). Predicting y or x values outside the range of observed values (extrapolation) is sometimes interesting, but it can easily yield ridiculous results.

Coefficient of determination

The total variation in y is obtained by squaring the deviations about the mean of the y variable ($\bar{y}$) and summing them up over all points.

The total variation can be decomposed into regression (explained) and residual (unexplained) components.



A best-fitting regression line is depicted; total, explained and unexplained deviations for a point are also depicted

The coefficient of determination, or r^2 , expresses the strength of the relationship between two variables.

It is the proportion of the variation in the y variable that is 'explained' by the variation in the x variable.

$$\begin{aligned} r^2 &= \frac{\text{Regression Sum of Squares}}{\text{Total Sum of Squares}} \quad \text{or} \\ &= 1 - \frac{\text{Residual Sum of Squares}}{\text{Total Sum of Squares}} \end{aligned} \quad (1)$$

r^2 can vary from 0 to 1. Values near 1 mean the variation in y can be explained by variation in x , and the y values fall on the regression line.

In other words, once x is known y can be predicted exactly, with no error or variability in the prediction. On the other hand, values near 0 mean there is very little or no relationship between x and y . Thus,

$$r^2 = 1 \quad \text{Regression SS} = \text{Total SS, or Residual SS} = 0$$

$$r^2 = 0 \quad \text{Residual SS} = \text{Total SS, or Regression SS} = 0$$

The square root of r^2 , with a negative sign if the slope is negative, is the Pearson correlation coefficient.

Assumptions and limitations

The use of correlation and regression depends on some underlying assumptions.

- ▶ The observations are assumed to be independent. For correlation both variables should be random variables, but for regression only the response variable y must be random and normally distributed.
- ▶ Both correlation and regression assume that the relationship between the two variables is linear. If this is not the case, data transformation may be helpful, or it may be necessary to use polynomial regression.

- ▶ When using a regression equation for prediction, errors in prediction may not be just random but also be due to inadequacies in the model. In particular, extrapolating beyond the range of the data is very risky.
- ▶ Linear regression assumes that the data points are independent of each other, i.e., the value of one data point does not depend on what the value of any other data point is.
- ▶ A phenomenon to be aware of that may arise with repeated measurements on individuals is regression to the mean.

MAT1106 Design and Analysis of Experiments

Design of Experiments

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Determining relationships between factors and responses

In an experimental set up, the relationships between input process variables (or factors) and output performance characteristics (responses) have to be determined.

For example, in a metal cutting operation, cutting speed, feed rate, type of coolant, and depth of cut can be treated as input variables, and surface finish of the finished part can be considered as an output performance characteristic.

Carrying out experiments

Experiments are carried out by varying input variables (or factors) and determining the impact on output functional performance (or response).

Objectives of experiments:

- ▶ Determine which set of factors in a process have the largest effects on performance.
- ▶ Determine optimum levels of these factors to obtain satisfactory performance.

A common experimental approach

A commonly used approach to determine relationships between input factors and output characteristics is the One-Factor-At-a-Time (OFAT) approach, where one factor is varied at a time keeping all other factors in the experiment fixed.

One-Factor-at-a-Time test sequence

- Stage 1 Fix levels of factors 2, 3, ..., k ; determine the optimal level of factor 1
- Stage 2 Use the optimal level of factor 1 from stage 1; fix levels of factors 3, 4, ..., k ; determine the optimal level of factor 2
- Stage 3 Use the optimal levels of factors 1, 2 from stages 1, 2; fix levels of factors 4, 5, ..., k ; determine optimal level of factor 3
- $\vdots$
- Stage k Use the optimal levels of factors 1, 2, ..., $k - 1$ from stages 1, 2, ..., $k - 1$; determine the optimal level of factor k

Drawbacks of the OFAT approach

The OFAT approach is time-consuming and resource-intensive.

If factors jointly, not just individually, influence the response, use of the OFAT approach will lead to incorrect or suboptimal results.

The need for a statistical approach

The problem of experiment efficiency is most acute when several factors must be investigated in an experiment. Time and cost efficiencies are also important objectives in experimental work.

Efficiency is achieved in statistically designed experiments because each observation generally provides information on all the factors of interest in the experiment.

Statistical methods play an important role in planning, conducting, analysing and interpreting data from experiments.

Benefits of using a statistical approach

- ▶ Better understanding of the relationship between key process inputs and outputs
- ▶ Reduced process design and development time
- ▶ Improved process yield
- ▶ Reduced process variability and hence better product performance consistency
- ▶ Reduced manufacturing costs
- ▶ Improved profits and return on investment

Design of Experiments was developed in the early 1920s by Sir Ronald Fisher at the Rothamsted Agricultural Field Research Station in London, England.

Fisher wanted to determine the effect of various fertilizers on different plots of land, while differentiating between other factors that can affect crop yield (such as soil characteristics and moisture content of the soil).

Fisher pioneered the use of DoE to differentiate between effect of fertilizer and effects of other factors.

What should be achieved using DoE?

A good experimental design should

- ▶ avoid systematic error
- ▶ be precise
- ▶ allow estimation of error
- ▶ have broad validity

Principles of experimental design

- ▶ Randomization
- ▶ Replication
- ▶ Blocking

Randomization

Instrument drift, power surges, equipment malfunctions, technician or operator errors, or a myriad of other undetectable influences can bias the results of an experiment.

Randomization protects against such unknown or unmeasured sources of possible experimental bias.

Randomization affords protection from bias by tending to average the bias effects over all levels of the factors in the experiment. When comparisons are made among levels of a factor, the bias effects will tend to cancel and the true factor effects will remain.

Statistical methods require that the observations (or errors) be independently distributed random variables. Randomization usually makes this assumption valid.

Uses of randomization

1. Protects against confounding.
2. Can form the basis for inference.

Randomization against confounding

Confounding occurs when the effect of one factor cannot be distinguished from that of another factor.

Consider planting corn variety A in Nashik and corn variety B in Ahmednagar. In this experiment, we cannot distinguish location effects from variety effects, as the variety factor and the location factor are confounded.

Randomization helps in reducing the error arising due to confounding.

Performing a randomization

Physical randomization is achieved via an actual physical act that produces random results with known properties.

Examples of physical randomization are

- ▶ coin tosses
- ▶ card draws from shuffled decks
- ▶ rolls of a die
- ▶ chits in a hat

Numerical randomization uses numbers taken from a table of random numbers or generated by a random number generator using software packages.

Question: How to randomly order n objects?

Answer: Take n chits numbered 1 through n and mix them well. The first object is given the number of the first chit drawn, and so on. The objects are then sorted so that their assigned numbers are in increasing order.

With good mixing, all orders of the objects are equally likely.

Replication

Replication entails an independent repeat run of each factor combination, and is not to be confused with repeated measurements.

Replication is important as it provides an estimate of the experimental error. This estimate of error becomes a basic unit of measurement for determining whether or not observed differences in the data are statistically significant.

Replication increases precision in the estimation of factor effects by reducing the standard deviations of the statistics used to estimate effects (due to increase in sample sizes).

Replication can result in a substantial increase in the time to conduct an experiment. Moreover, if the material is expensive, replication may lead to exorbitant material costs.

Blocking

Blocking is a design technique used to improve the precision with which comparisons among the factors of interest are made.

Blocking increases precision (decreases variability) in the estimation of factor effects by controlling the systematic variation attributable to nonhomogeneous experimental units or test conditions.

Batch-to-batch, day-to-day, supplier-to-supplier variations can be controlled for by grouping similar experimental runs into blocks. Blocks can also be batches of raw materials, different operators, different vendors, etc.

A block is a set of relatively homogeneous experimental conditions. Observations collected under the same experimental conditions are said to be in the same block.

Two batches of raw materials are required to conduct a set of experimental runs.

To eliminate batch-to-batch material variability, each batch of raw material can be treated as a single block, because the variability within a batch would be expected to be smaller than the variability between batches.

MAT1106 Design and Analysis of Experiments

Design of Experiments

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Factorial experiments

Experiments are conducted to investigate the effects of one or more factors on a response. When an experiment consists of two or more factors, the factors can influence the response individually or jointly.

Often, as in the case of one-factor-at-a-time experimentation, the experimental design does not allow one to properly assess the joint effects of the factors.

Factorial experiments, which include all possible factor-level combinations in the experimental design, allow joint effects of factors to be estimated.

Completely randomized design

Factorial experiments can be conducted in a wide variety of experimental designs.

One of the most straightforward designs to implement is the completely randomized design.

In a completely randomized design, all the factor-level combinations in the experiment are randomly assigned to experimental units, if appropriate, or to the sequence of test runs.

Construction of factorial experiments in completely randomized designs

1. Enumerate all factor-level combinations. Include repeat tests.
2. Number the factor-level combinations, including any repeat tests, sequentially from 1 to n .
3. From a random-number table or from a computer-generated sequence of random numbers, obtain a random sequence of the integers 1 to n .
4. Assign the factor-level combinations to the experimental units or conduct the test runs in the order specified by the random number sequence. If both a randomized assignment to experimental units and a randomized test sequence are to be included in the design, use two random number sequences.

Including repeat runs in experiments

Experimental designs should, whenever possible, include repeat test runs.

The inclusion of repeat tests in an experimental design does not necessarily require that each factor-level combination be repeated or that each be repeated an equal number of times.

The only requirement is that a sufficient number of repeat tests be included so that a satisfactory estimate of experimental error can be obtained.

In general, it is unwise to select a single factor-level combination and repeat it several times. A better approach when all combinations cannot be repeated in an experiment is to randomly select, without replacement, the factor-level combinations to be repeated.

As a rough rule of thumb, a minimum of six repeat tests should be included to estimate the experimental error standard deviation.

Interactions

An interaction exists among two or more factors if the effect of one factor on a response depends on the levels of other factors.

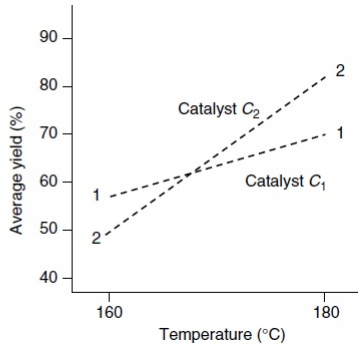
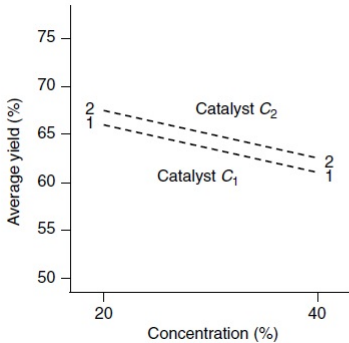
When interactions occur, the factors involved cannot be evaluated individually.

Factorial experiments conducted in completely randomized designs are especially useful for evaluating joint factor effects (interactions).

A chemical process was carried out, and the yield calculated.

The factors of interest in this study were the operating temperature of the plant, the type of catalyst used in the process, and the concentration of the catalyst.

The average yields for each catalyst concentration and each operating temperature were plotted separately for each catalyst.



Interaction plots for chemical yield data. Dashed rather than solid lines are used, as the relationships are not necessarily linear.

Catalyst concentration has approximately the same effect on yield for each catalyst, and vice versa. Therefore, there is no two-factor interaction between catalyst and concentration.

On the other hand, the effect on yield of operating temperature depends on which catalyst is used. In other words, the yield depends on both the temperature and the catalyst. Therefore, there is a two-factor interaction between catalyst and temperature.

Graphical visualization of interaction

Graphical procedures allow a visual inspection of factor effects. In general, interactions are indicated graphically by nonparallel trends in plots of average responses.

The dashed line segments connecting the average responses for each level of the factors need not intersect. Any nonparallel changes in the average responses is an indication of an interaction.

As with all statistical analyses, the existence (or absence) of interaction effects implied by graphical displays should be confirmed with formal statistical inference procedures that account for experimental variation.

Calculation of factor effects

An effect is a change in the average response corresponding to a change in factor-level combinations.

The calculation of effects is facilitated by the introduction of algebraic notation for individual responses and averages of responses.

Notation for responses

Factors in designed experiments are denoted by uppercase Latin letters, $A, B, C, \dots, K$.

An individual response is represented by a lowercase Latin letter, usually y , having one or more subscripts. Subscripts, one for each factor, designate the specific factor-level combination for which the response is obtained.

There may also be one or more subscripts designating repeat observations for fixed factor levels.

Factor	Factor symbol	Level	Subscript symbol
Alloy	A	Steel	$i = 1$
		Aluminum	$i = 2$
Sleeve	B	Porous	$j = 1$
		Nonporous	$j = 2$
Lubricant	C	Lub 1	$k = 1$
		Lub 2	$k = 2$
		Lub 3	$k = 3$
		Lub 4	$k = 4$

A fourth subscript l could be used to denote the repeat-test number, provided there is at least one repeat test; otherwise, the fourth subscript is superfluous and not included.

With this convention in notation, responses from a three-factor experiment with r repeat tests per factor-level combination would be denoted y_{ijkl} with $i = 1, \dots, a$, $j = 1, \dots, b$, $k = 1, \dots, c$, and $l = 1, \dots, r$.

The upper limits a , b , c , and r on the subscripts denote the numbers of levels of the three factors A , B , C and the number of repeat observations, respectively.

Average responses are represented by replacing one or more of the subscripts by a dot and placing a bar over the symbol for the response. Using dot notation,

$$\bar{y}_{\bullet\bullet\bullet} = n^{-1} \sum_{ijkl} y_{ijkl}$$

denotes the overall average response for a three-factor experiment with repeat tests, where $n = abcr$ and the summation is over all the observations (all possible values of the subscripts).

Similarly,

$$\bar{y}_{i\bullet\bullet\bullet} = (bcr)^{-1} \sum_{jkl} y_{ijkl}$$

is the average response for all responses having the i th level of factor A .

The symbol $\bar{y}_{2\bullet 1\bullet}$ denotes the average response for all test runs having the second level of the first factor and the first level of the third factor.

The symbol $\bar{y}_{ijk\bullet}$ denotes a typical average response, across repeat observations, for one of the factor-level combinations.

Effects representation

Consider a factorial experiment with each factor having two levels and r repeat observations for each combination of levels of the factors.

One way to represent factor-level combinations is using the 'effects representation'.

Let one of the levels of a factor be coded -1 and the other level be coded $+1$. It is arbitrary which level receives each code, although it is customary for quantitative factors to let the lower level be -1 .

Factor levels		Effects representation	
Factor <i>A</i>	Factor <i>B</i>	<i>A</i>	<i>B</i>
Lower	Lower	-1	-1
Lower	Upper	-1	+1
Upper	Lower	+1	-1
Upper	Upper	+1	+1

Effects			Average response
<i>A</i>	<i>B</i>	<i>AB</i>	
-1	-1	+1	$\bar{y}_{11\bullet}$
-1	+1	-1	$\bar{y}_{12\bullet}$
+1	-1	-1	$\bar{y}_{21\bullet}$
+1	+1	+1	$\bar{y}_{22\bullet}$

Effects coding of factor levels

1. Designate one level of each factor as -1 and the other level as $+1$.
2. Make a table with column headings for each of the factors $A, B, C, \dots, K$.
3. Let $n = 2^k$, where k is the number of factors.
4. Set the first $n/2$ of the levels for factor A equal to -1 and the last $n/2$ equal to $+1$. Set the first $n/4$ levels of factor B equal to -1 , the next $n/4$ equal to $+1$, the next $n/4$ equal to -1 , and the last $n/4$ equal to $+1$. Set the first $n/8$ of the levels for factor C equal to -1 , the next $n/8$ equal to $+1$, etc. Continue in this fashion until the last column (for factor K) has alternating -1 and $+1$ signs.

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Design of Experiments

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Main effects and interaction effects

For a two-factor, two-level experiment with repeat tests, the main effect for factor A , designated $M(A)$, is calculated as follows:

$$M(A) = \bar{y}_{2\bullet\bullet} - \bar{y}_{1\bullet\bullet}$$

Similarly, the main effect of factor B is calculated as follows:

$$M(B) = \bar{y}_{\bullet 2\bullet} - \bar{y}_{\bullet 1\bullet}$$

The main effect for factor A at each level of factor B is calculated as follows for the lower level of factor B :

$$M(A)_{j=1} = \bar{y}_{21\bullet} - \bar{y}_{11\bullet}$$

The main effect for factor A at each level of factor B is calculated as follows for the upper level of factor B :

$$M(A)_{j=2} = \bar{y}_{22\bullet} - \bar{y}_{12\bullet}$$

From these effects the interaction between factors A and B , denoted $I(AB)$, can be calculated as follows:

$$I(AB) = M(A)_{j=2} - M(A)_{j=1}$$

Calculation of factor effects for two-level factors

1. Construct the effects representation for each main effect and each interaction.
2. Calculate linear combinations of the average responses (or individual responses if $r = 1$), using the signs in the effects column for each main effect and for each interaction.
3. Divide the respective linear combinations of average responses by 2^{k-1} , where k is the number of factors in the experiment.

This procedure can be used with any number of factors so long as there are an equal number of repeat observations for each of the factor-level combinations included in the experiment (that is, the design is balanced).

Note that the effect is quantified as the change in response over half the range of the factor. For example, if temperature is tested at two levels, say 375 K and 425 K, an effect of -5 represents a reduction in y of 5 units for every increase of 25 K.

If repeat runs are carried out, the standard error of any of the effects can be estimated using the following formula:

$$SE = \frac{2 \times SD}{\sqrt{n}}$$

The standard error is multiplied by 2 to take into account the change in response from the lower level to the upper level (double the distance).

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April 8, 2021



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Calculation of factor effects for factors with more than two levels

The general procedure for representing effects of two-level factors as differences of averages can be extended to any number of factor levels.

A main effect for a two-level factor can be uniquely defined as the difference between the average responses at the two levels of the factor. When a factor has more than two levels, say m , there is more than one main effect and there are many ways of defining the main effects.

One way is to calculate differences between averages for levels 1 to $m - 1$ from the average for level m :

$$\bar{y}_{1\bullet} - \bar{y}_{m\bullet}, \bar{y}_{2\bullet} - \bar{y}_{m\bullet}, \dots, \bar{y}_{m-1\bullet} - \bar{y}_{m\bullet}$$

An alternative way is to calculate the differences of the averages at the various levels from the overall average:

$$\bar{y}_{1\bullet} - \bar{y}_{\bullet\bullet}, \bar{y}_{2\bullet} - \bar{y}_{\bullet\bullet}, \dots, \bar{y}_{m\bullet} - \bar{y}_{\bullet\bullet}$$

The main effects can be determined by taking linear combinations of the effects.

Interactions and factorial experiments

Presence of interactions requires that factors be evaluated jointly rather than individually.

Experiments should be designed to measure interactions. Failure to do so can lead to misleading, even incorrect, conclusions.

Factorial experiments enable all joint factor effects to be estimated.

Fractional factorial experiments

Because of the possible existence of interactions, complete factorial experiments should be conducted whenever possible.

In experiments with a large number of factors, however, complete factorial experiments may not be economically feasible even if it is believed that some interactions among the factors may exist.

In such settings, fractional factorial and screening experiments are important alternatives.

Fractional factorial experiments are important alternatives to complete factorial experiments when budgetary, time, or experimental constraints preclude the execution of complete factorial experiments.

In addition, experiments that involve many factors are routinely conducted as fractional factorials because it is not necessary to test all possible factor-level combinations to estimate the important factor effects.

Highly fractionated designs used for screening experiments are often used to obtain information about only the major factor effects on a response.

Confounding of factor effects

Whenever fractional factorial experiments are conducted, some effects are confounded with one another.

The goal in the design of fractional factorial experiments is to ensure that the effects of primary interest are either unconfounded with other effects or, if that is not possible, confounded with effects that are not likely to have appreciable magnitudes.

The efficient design of fractional factorial experiments utilizes the planned confounding of factor effects in their construction.

Half-fractional design

A half-fractional design consists of $\frac{1}{2}2^k = 2^{k-1}$ runs.

For example, a two-level, three-factor experiment that is to be run as a half-fractional experiment will have $2^{3-1} = 2^2 = 4$ runs, compared with $2^3 = 8$ runs for a full factorial experiment.

Designing half-factorial experiments

1. The effects representation of the first $k - 1$ factors are constructed.
2. The effects representation of the final k^{th} factor is generated from the product of the previous $k - 1$ factors. This is called the 'generating relation' for the design.

A	B	$C = AB$	AC	BC	ABC
-1	-1	+1	-1	-1	+1
-1	+1	-1	+1	-1	+1
+1	-1	-1	-1	+1	+1
+1	+1	+1	+1	+1	+1

Note that $A = BC$, $B = AC$ and $C = AB$ (this last association is intentional). Thus, the main effects and two-factor interactions are confounded.

In other words, A is an alias for BC , B is an alias for AC and C is an alias for AB , as these aliased effects cannot be separated.

The complementary half-fraction can be generated by $C = -AB$.

In fractional factorial experiments, planned confounding of effects is carried out so that the effects of most interest are either unconfounded or only confounded with effects that are believed to be negligible relative to the uncontrolled experimental error.

Highly fractionated designs

The purpose of a fractionated design is to reduce the number of experiments due to, for example, budgetary or time constraints that do not allow carrying out a full factorial experiment.

A full factorial experiment is often not required, especially when k is greater than about 4, since the higher-order interaction terms are almost always insignificant.

In general, a 2^{k-p} fractional factorial design can be used instead of a full factorial design.

An experimental design with 2^{k-1} runs is called a half-fraction, while a design with 2^{k-2} runs is called a quarter-fraction, and so on.

Design generator

A fractional factorial experiment is designed by confounding some effects with others.

For example, a half-factorial experiment involving three factors at two levels can be designed by confounding (or aliasing) one main effect with a two-factor interaction effect.

Here, the 'design generator' is represented by $C = AB$.

The 'defining relation' of this design is

$$C \times C = C \times AB$$

$$C^2 = ABC = I$$

where I is the identity element.

A	B	$C = AB$	AC	BC	ABC	I
-1	-1	+1	-1	-1	+1	+1
-1	+1	-1	+1	-1	+1	+1
+1	-1	-1	-1	+1	+1	+1
+1	+1	+1	+1	+1	+1	+1

The defining relation for this design is $I = ABC$. The term ABC is called the 'generator' of this design.

Note that the defining relation for the alternate or complementary half-fractional design is $I = -ABC$.

Using the defining relation of a design, the alias structure for that particular design can be generated.

For example, if the defining relation is $I = ABC$, the alias of A can be determined as follows:

$$A \times I = A \times ABC$$

$$A = A^2BC \quad (\because I = 1)$$

$$A = BC \quad (\because A^2 = I)$$

Aliases of other factors can be similarly determined.

The generator term ABC is called a 'word'.

As a rule, a 2^{k-p} factorial design is produced by p generators and has a defining relation of 2^p words.

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Design resolution

An important guide that is used in selecting fractional factorial experiments is the concept of 'design resolution'.

Design resolution is defined in terms of which effects are unconfounded with which other effects.

Designs of resolution III, IV, and V are particularly important.

Resolution III designs: These are designs in which no main effects are aliased with any other main effect, but main effects are aliased with two-factor interactions and some two-factor interactions may be aliased with each other.

A 2^{3-1} design with $I = ABC$ is a resolution III design (2_{III}^{3-1}).

Resolution IV designs: These are designs in which no main effect is aliased with any other main effect or with any two-factor interaction, but two-factor interactions are aliased with each other.

A 2^{4-1} design with $I = ABCD$ is a resolution IV design (2_{IV}^{4-1}).

Resolution V designs: These are designs in which no main effect or two-factor interaction is aliased with any other main effect or two-factor interaction, but two-factor interactions are aliased with three-factor interactions.

A 2^{5-1} design with $I = ABCDE$ is a resolution V design (2_V^{5-1}).

In general, the resolution of a two-level fractional factorial design is equal to the number of letters in the shortest 'word' in the defining relation.

Fractional designs that have the highest possible resolution consistent with the degree of fractionation required should be employed.

The higher the resolution, the less restrictive the assumptions that are required regarding which interactions are negligible to obtain a unique interpretation of the results.

If it is believed that only main effects are present and all interactions are negligible, a resolution III design is sufficient for estimating main effects.

If it is believed that some two-factor interactions may be non-negligible but all three-factor and higher interactions are negligible, a resolution IV design is sufficient for estimating main effects.

Low-resolution fractional factorials are often used as screening designs, where many factors are screened to see if any of them has an effect.

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Plackett-Burman designs

A special class of two-level fractional factorial experiments that is widely used in screening experiments was proposed by Plackett and Burman.

These experiments have resolution III when conducted in completely randomized designs and are often referred to as 'Plackett-Burman' designs.

Plackett-Burman designs are available for experiments that have the number of test runs equal to a multiple of four.

These designs are very useful for economically detecting large main effects, assuming all interactions are negligible when compared with the few important main effects.

Plackett-Burman design for 12 runs and 11 two-level factors

Run	A	B	C	D	E	F	G	H	I	J	K
1	+	+	+	+	+	+	+	+	+	+	+
2	-	+	-	+	+	+	-	-	-	+	-
3	-	-	+	-	+	+	+	-	-	-	+
4	+	-	-	+	-	+	+	+	-	-	-
5	-	+	-	-	+	-	+	+	+	-	-
6	-	-	+	-	-	+	-	+	+	+	-
7	-	-	-	+	-	-	+	-	+	+	+
8	+	-	-	-	+	-	-	+	-	+	+
9	+	+	-	-	-	+	-	-	+	-	+
10	+	+	+	-	-	-	+	-	-	+	-
11	-	+	+	+	-	-	-	+	-	-	+
12	+	-	+	+	+	-	-	-	+	-	-

The response surface

A response variable can be expressed as an empirical function of one or more quantitative factors. A response function can be used to model the relationship:

$$y = f(x_1, x_2, \dots, x_k)$$

where y is the response and $x_1, x_2, \dots, x_k$ are quantitative levels of the factors of interest.

The form of the function f can be found by fitting models to data obtained from designed experiments.

The form of the function is used to summarize results of the experiment and to predict the response for values of the quantitative factors.

The function f defines the 'response surface', which is the geometric representation obtained when a response variable is plotted as a function of one or more quantitative factors.

Uses of response surface

Response surface is used to determine

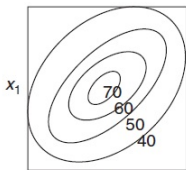
- ▶ sensitivity of the response to the factors of interest
- ▶ factor levels for which the response variable is optimum

Contour plots

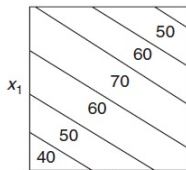
A response surface can have various shapes, depending on the form of the response function.

For example, a two-factor response surface can be plotted as contours of constant response.

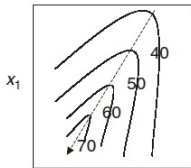
A contour plot is a series of lines or curves that identify values of the factors for which the response is constant. Curves for several (usually equally spaced) values of the response are plotted.



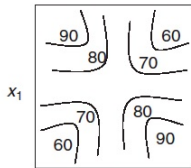
(a)



(b)



(c)



(d)

Response-surface contours in two factors: (a) 'mound-shaped' maximum, (b) stationary ridge, (c) rising ridge, and (d) saddle

Designs for fitting response surfaces

Complete and fractional factorial experiments in completely randomized designs are extremely useful in exploring the factor space to identify the region where the optimum response is located.

Three-level factorial experiments are often conducted to fit response surfaces.

Typical factor levels for a 3^2 factorial design

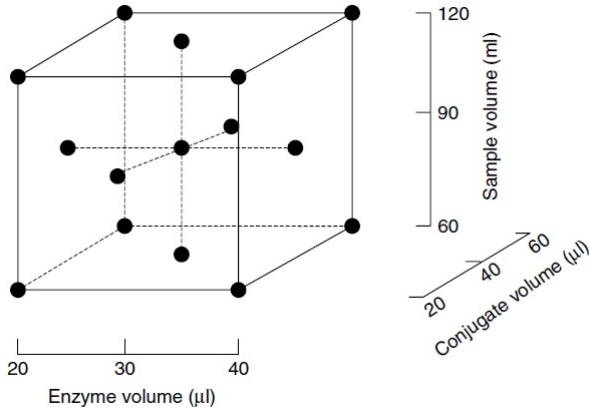
Factor levels		Effects representation	
Factor <i>A</i>	Factor <i>B</i>	<i>A</i>	<i>B</i>
Low	Low	−1	−1
Low	Middle	−1	0
Low	High	−1	+1
Middle	Low	0	−1
Middle	Middle	0	0
Middle	High	0	+1
High	Low	+1	−1
High	Middle	+1	0
High	High	+1	+1

Central composite design and Box-Behnken design are fractions of the 3^k factorials that only require enough observations to estimate the second-order effects of the response surface.

Central composite design

The total number of test runs in a central composite design based on a complete 2^k factorial is $n = 2^k + 2k + m$, where m are repeat observations.

This count usually is less than 3^k , so that fewer observations are required than in a 3^k factorial.

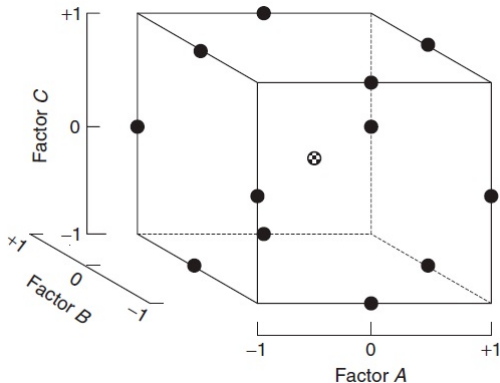


Face-centred central composite design for a study involving
three factors at three levels
 $n = 8 + 6 + 3 = 17 (< 3^3 = 27)$

Box-Behnken design

The Box-Behnken design is another alternative to the 3^k factorial.

The designs are formed by combining 2^k factorials with incomplete block designs.



Box-Behnken design for a study involving
three factors at three levels
 $n = 12 + 3 = 15 (< 3^3 = 27)$

<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>	<i>I</i>	No. of runs
±1	0	0	±1	0	0	±1	0	0	8
0	±1	0	0	±1	0	0	±1	0	8
0	0	±1	0	0	±1	0	0	±1	8
±1	±1	±1	0	0	0	0	0	0	8
0	0	0	±1	±1	±1	0	0	0	8
0	0	0	0	0	0	±1	±1	±1	8
±1	0	0	0	±1	0	0	0	±1	8
0	0	±1	±1	0	0	0	±1	0	8
0	±1	0	0	0	±1	±1	0	0	8
±1	0	0	0	0	±1	0	±1	0	8
0	±1	0	±1	0	0	0	0	±1	8
0	0	±1	0	±1	0	±1	0	0	8
±1	0	0	±1	0	0	±1	0	0	8
0	±1	0	0	±1	0	0	±1	0	8
0	0	±1	0	0	±1	0	0	±1	8
0	0	0	0	0	0	0	0	0	10
									130

Box-Behnken design for a study involving
 nine factors at three levels
 $n = 130 (< 3^9 = 19683)$

Preferred properties of response surface designs

- ▶ Result in a good fit of the model to the data.
- ▶ Provide an estimate of 'pure' experimental error.
- ▶ Be robust to the presence of outliers in the data.
- ▶ Be robust to errors in control of design levels.
- ▶ Be cost effective.
- ▶ Allow experiments to be done in blocks.

Normal Probability Table

Z-Score	0.09	0.08	0.07	0.06	0.05	0.04	0.03	0.02	0.01	0.00
-3.9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
-3.8	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
-3.7	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
-3.6	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0002	0.0002
-3.5	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002
-3.4	0.0002	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003
-3.3	0.0003	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0005	0.0005	0.0005
-3.2	0.0005	0.0005	0.0005	0.0006	0.0006	0.0006	0.0006	0.0006	0.0007	0.0007
-3.1	0.0007	0.0007	0.0008	0.0008	0.0008	0.0008	0.0009	0.0009	0.0009	0.0010
-3.0	0.0010	0.0010	0.0011	0.0011	0.0011	0.0012	0.0012	0.0013	0.0013	0.0013
-2.9	0.0014	0.0014	0.0015	0.0015	0.0016	0.0016	0.0017	0.0018	0.0018	0.0019
-2.8	0.0019	0.0020	0.0021	0.0021	0.0022	0.0023	0.0023	0.0024	0.0025	0.0026
-2.7	0.0026	0.0027	0.0028	0.0029	0.0030	0.0031	0.0032	0.0033	0.0034	0.0035
-2.6	0.0036	0.0037	0.0038	0.0039	0.0040	0.0041	0.0043	0.0044	0.0045	0.0047
-2.5	0.0048	0.0049	0.0051	0.0052	0.0054	0.0055	0.0057	0.0059	0.0060	0.0062
-2.4	0.0064	0.0066	0.0068	0.0069	0.0071	0.0073	0.0075	0.0078	0.0080	0.0082
-2.3	0.0084	0.0087	0.0089	0.0091	0.0094	0.0096	0.0099	0.0102	0.0104	0.0107
-2.2	0.0110	0.0113	0.0116	0.0119	0.0122	0.0125	0.0129	0.0132	0.0136	0.0139
-2.1	0.0143	0.0146	0.0150	0.0154	0.0158	0.0162	0.0166	0.0170	0.0174	0.0179
-2.0	0.0183	0.0188	0.0192	0.0197	0.0202	0.0207	0.0212	0.0217	0.0222	0.0228
-1.9	0.0233	0.0239	0.0244	0.0250	0.0256	0.0262	0.0268	0.0274	0.0281	0.0287
-1.8	0.0294	0.0301	0.0307	0.0314	0.0322	0.0329	0.0336	0.0344	0.0351	0.0359
-1.7	0.0367	0.0375	0.0384	0.0392	0.0401	0.0409	0.0418	0.0427	0.0436	0.0446
-1.6	0.0455	0.0465	0.0475	0.0485	0.0495	0.0505	0.0516	0.0526	0.0537	0.0548
-1.5	0.0559	0.0571	0.0582	0.0594	0.0606	0.0618	0.0630	0.0643	0.0655	0.0668
-1.4	0.0681	0.0694	0.0708	0.0721	0.0735	0.0749	0.0764	0.0778	0.0793	0.0808
-1.3	0.0823	0.0838	0.0853	0.0869	0.0885	0.0901	0.0918	0.0934	0.0951	0.0968
-1.2	0.0985	0.1003	0.1020	0.1038	0.1056	0.1075	0.1093	0.1112	0.1131	0.1151
-1.1	0.1170	0.1190	0.1210	0.1230	0.1251	0.1271	0.1292	0.1314	0.1335	0.1357
-1.0	0.1379	0.1401	0.1423	0.1446	0.1469	0.1492	0.1515	0.1539	0.1562	0.1587
-0.9	0.1611	0.1635	0.1660	0.1685	0.1711	0.1736	0.1762	0.1788	0.1814	0.1841
-0.8	0.1867	0.1894	0.1922	0.1949	0.1977	0.2005	0.2033	0.2061	0.2090	0.2119
-0.7	0.2148	0.2177	0.2206	0.2236	0.2266	0.2296	0.2327	0.2358	0.2389	0.2420
-0.6	0.2451	0.2483	0.2514	0.2546	0.2578	0.2611	0.2643	0.2676	0.2709	0.2743
-0.5	0.2776	0.2810	0.2843	0.2877	0.2912	0.2946	0.2981	0.3015	0.3050	0.3085
-0.4	0.3121	0.3156	0.3192	0.3228	0.3264	0.3300	0.3336	0.3372	0.3409	0.3446
-0.3	0.3483	0.3520	0.3557	0.3594	0.3632	0.3669	0.3707	0.3745	0.3783	0.3821
-0.2	0.3859	0.3897	0.3936	0.3974	0.4013	0.4052	0.4090	0.4129	0.4168	0.4207
-0.1	0.4247	0.4286	0.4325	0.4364	0.4404	0.4443	0.4483	0.4522	0.4562	0.4602
-0.0	0.4641	0.4681	0.4721	0.4761	0.4801	0.4840	0.4880	0.4920	0.4960	0.5000

[illegible]